

Factors, Multiples, GCF & LCM

Name: _____ Date: _____ Class: _____

Part A · Factors, Prime Factorization & GCF

◆ Worked Examples

- Factors of 20: 1, 2, 4, 5, 10, 20 (numbers that divide 20 evenly)
- Prime factorization of 24: $2 \times 2 \times 2 \times 3$ (only primes multiplied together)
- GCF(8, 12): factors of 8 = {1, 2, 4, 8}; factors of 12 = {1, 2, 3, 4, 6, 12} $\rightarrow$ GCF = 4

1. List all factors of 12: _____
2. List all factors of 18: _____
3. List all factors of 24: _____
4. List all factors of 40: _____
5. Prime factorization of 36 (use $\times$ between primes): _____
6. Prime factorization of 60 (use $\times$ between primes): _____
7. GCF (12, 18) = _____
8. GCF (20, 30) = _____
9. GCF (24, 36) = _____
10. Word problem — Emma has 15 stickers and 20 pencils. She wants to make identical gift bags with no leftovers. What is the LARGEST number of bags she can make?
Answer: _____ bags

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Part B · Multiples & LCM

◆ Worked Examples

- First 5 multiples of 3: 3, 6, 9, 12, 15 (count by 3s)
- LCM(4, 6): multiples of 4 = 4, 8, 12, 16; multiples of 6 = 6, 12, 18 → LCM = 12

11. First 5 multiples of 4 (starting at 4): _____

12. First 5 multiples of 6 (starting at 6): _____

13. First 5 multiples of 7 (starting at 7): _____

14. First 5 multiples of 9 (starting at 9): _____

15. LCM (3, 4) = _____

16. LCM (4, 6) = _____

17. LCM (5, 8) = _____

18. LCM (6, 9) = _____

19. Word problem — Two lighthouses blink. One blinks every 6 seconds; the other every 8 seconds. If they blink together at time zero, when do they NEXT blink together?

Answer: _____ seconds

20. Word problem — Bus A leaves the station every 12 minutes; Bus B every 15 minutes. Both just left at noon. When will they NEXT leave together?

Answer: _____ minutes after noon

Factors, Multiples, GCF & LCM — Answer Key

Part A · Factors, Prime Factorization & GCF

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|---------------------------------------|--|
| 1. Factors of 12: | 1, 2, 3, 4, 6, 12 |
| 2. Factors of 18: | 1, 2, 3, 6, 9, 18 |
| 3. Factors of 24: | 1, 2, 3, 4, 6, 8, 12, 24 |
| 4. Factors of 40: | 1, 2, 4, 5, 8, 10, 20, 40 |
| 5. Prime factorization of 36: | $2 \times 2 \times 3 \times 3$ (or $2^2 \times 3^2$) |
| 6. Prime factorization of 60: | $2 \times 2 \times 3 \times 5$ (or $2^2 \times 3 \times 5$) |
| 7. GCF (12, 18): | 6 |
| 8. GCF (20, 30): | 10 |
| 9. GCF (24, 36): | 12 |
| 10. Emma's gift bags (GCF of 15, 20): | 5 bags (each bag: 3 stickers + 4 pencils) |

Part B · Multiples & LCM

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|--------------------------------|----------------------|
| 11. First 5 multiples of 4: | 4, 8, 12, 16, 20 |
| 12. First 5 multiples of 6: | 6, 12, 18, 24, 30 |
| 13. First 5 multiples of 7: | 7, 14, 21, 28, 35 |
| 14. First 5 multiples of 9: | 9, 18, 27, 36, 45 |
| 15. LCM (3, 4): | 12 |
| 16. LCM (4, 6): | 12 |
| 17. LCM (5, 8): | 40 |
| 18. LCM (6, 9): | 18 |
| 19. Lighthouses (LCM of 6, 8): | 24 seconds |
| 20. Buses (LCM of 12, 15): | 60 minutes (1:00 PM) |